

Algorithm 1: Unified State Update Procedure

Input

ρ, α, t



Phase 1

C2: Timeout
Enforcement



Phase 2

C1: Risk-Based
Escalation



Phase 3

C4: Human
Approval Gate



Phase 4

C5: Controlled
De-escalation



Phase 5

C3: Audit
Logging



Output

$s', \text{log entry}$